

PS26189 — AI-Powered Criminal Network Analysis System

Final 6-Member Development Workflow — Dataset-Aligned & Performance-Optimized

Problem Statement ID: 26189

Organization: Ministry of Home Affairs — National Crime Records Bureau (NCRB), Women Safety Division

Theme: Blockchain & Cybersecurity **Category:** Software

Purpose of this document

This document converts the team's existing 6-member development plan into one concrete, dataset-aligned workflow for PS26189. The division of work remains: Anika — Backend + Database; Krisha — Frontend + Dashboard; Aayushman — Core Intelligence / Detection Engine; Aayush — Data & Evidence Processing; Safina — Cybersecurity + Testing / QA; Meet — Architecture + Integration + AI + Final Demo.

Dataset basis: the supplied FIR dataset contains 544 FIR images and 2,447 annotated OCR/text records. The records include fields such as image identifier/name, bounding-box information, confidence/score, category ID and extracted text. Therefore, the system should use the dataset as the evidence-ingestion and text-extraction foundation, then derive entities, relationships and network intelligence from the processed FIR content.

1. COMPLETE SYSTEM WORKFLOW

FIR DATASET → AAYUSH DATA PROCESSING → STRUCTURED FIR JSON → AAYUSHMAN INTELLIGENCE ENGINE → ANALYSIS JSON → ANIKA BACKEND + DATABASE → KRISHA INVESTIGATION DASHBOARD → MEET AI EXPLANATION / INTEGRATION → INVESTIGATOR REPORT

Safina's Security + QA layer operates across all stages rather than only at the end.

Detailed pipeline:

1. FIR images / OCR annotations
2. OCR text reconstruction and cleaning
3. Validation, normalization and deduplication
4. Entity extraction
5. Entity resolution / linking
6. Relationship extraction
7. Knowledge/network graph construction
8. Graph analytics and suspicious-pattern detection
9. Explainable risk scoring
10. Alerts and investigator insights
11. Backend persistence and APIs
12. Interactive investigation dashboard
13. AI-generated explanation based only on detected evidence
14. Investigation report

2. DATASET → SYSTEM MAPPING

Dataset / Input	System Use
FIR image	Evidence source and visual document reference
image_id / image_name	Unique FIR/evidence linkage
OCR/text	Primary unstructured evidence for information extraction
bbox	Location of detected text in the source image; useful for evidence traceability
score/confidence	OCR/detection confidence; useful for filtering or flagging uncertain text
category_id	Annotation classification information retained during preprocessing

Important: the dataset should not be presented as if it already contains a complete criminal relationship database. The project derives relationships and network intelligence from the FIR evidence through processing and analysis.

3. AAYUSH — DATA & EVIDENCE PROCESSING

Goal: convert the supplied FIR/OCR dataset into clean, structured and traceable evidence.

Pipeline:

FIR Dataset → Dataset Loader → OCR/Text Reconstruction → Cleaning → Normalization → Entity Extraction → Relationship Candidates → Deduplication → Structured FIR JSON

Suggested module structure:

```
processing/  
■■■ dataset_loader.py  
■■■ ocr_parser.py  
■■■ text_cleaner.py  
■■■ normalizer.py  
■■■ entity_extractor.py  
■■■ relationship_extractor.py  
■■■ deduplicator.py  
■■■ schema.py
```

Entities to detect when evidence exists:

PERSON • LOCATION • ORGANIZATION • PHONE • VEHICLE • DATE • TIME • FIR/CASE NUMBER • ADDRESS • CRIME TYPE • MONEY/AMOUNT • DOCUMENT REFERENCE

Data-quality handling:

Missing values • duplicate OCR records • duplicate entities • invalid records • inconsistent dates • inconsistent naming
• OCR noise • low-confidence text

Output example:

```
{ "fir_id": "FIR_001", "source_image": "image_name", "text": "...", "entities": [  
  {"id": "P001", "type": "PERSON", "value": "..."}, {"id": "L001", "type": "LOCATION", "value": "..."} ],  
  "relationships": [] }
```

4. AAYUSHMAN — CORE INTELLIGENCE / DETECTION ENGINE

Goal: turn structured FIR evidence into explainable network intelligence.

Pipeline:

Structured FIR → Entity Resolution → Relationship Mining → Graph Construction → Graph Analytics → Pattern Detection → Risk Scoring → Suspicious Clusters → Alerts

Core graph entities:

Person, Phone, Email, Account, Organization, Location, Device, IP, Transaction, Event — only where supported by the processed evidence.

Relationship examples:

Person → uses → Phone
Person → connected_to → Person
Person → visited → Location
Person → owns → Account
Account → transaction → Account

Performance-oriented intelligence approach:

1. Rule/regex-based extraction for structured identifiers such as phone numbers, dates, amounts and case references.
2. Entity normalization and fuzzy/contextual matching for candidate entity resolution.
3. Graph analytics such as degree centrality, betweenness centrality, PageRank, connected components and community detection.
4. Pattern detection for repeated identifiers, cross-FIR appearances, dense clusters and unusual connection structures.
5. Explainable scoring instead of opaque/random scores.

Risk score example:

Risk Score: 87/100 – HIGH Reasons: • Connected to multiple suspicious/high-risk entities • Appears across multiple FIRs • Shares an identifier with other entities • Located in a high-density network cluster

5. ANIKA — BACKEND + DATABASE

Goal: provide the central API and persistent data layer connecting processing, intelligence and the dashboard.

Recommended database structure:

USERS ■■■■ CASES ■■■■ FIRS ■■■■ EVIDENCE ■■■■ ENTITIES ■ ■■■■ ENTITY_IDENTIFIERS ■■■■ RELATIONSHIPS ■■■■
ANALYSIS_RESULTS ■■■■ RISK_SCORES ■■■■ ALERTS ■■■■ INVESTIGATIONS ■■■■ REPORTS

Core APIs:

POST /api/auth/login
POST /api/cases
GET /api/cases
GET /api/cases/{id}
POST /api/evidence/upload
GET /api/evidence/{id}
GET /api/entities
GET /api/relationships
POST /api/analyze
GET /api/alerts
GET /api/reports

Critical relationship schema:

relationship_id • source_entity_id • target_entity_id • relationship_type • confidence • source_fir_id • evidence_id • created_at

Why source_fir_id and evidence_id matter:

Every important connection shown to an investigator should be traceable back to the FIR/evidence that produced it. This supports explainability and reduces the risk of presenting an unsupported connection.

6. KRISHA — FRONTEND + INVESTIGATION DASHBOARD

Goal: make the intelligence understandable to an investigator within seconds.

Main navigation:

LOGIN → DASHBOARD → CASES → CASE DETAILS → INVESTIGATION → EVIDENCE → NETWORK GRAPH → ALERTS → REPORT

Main investigation screen should show:

Network graph • Case/FIR context • Risk level • Entity count • Relationship count • Alerts • Evidence references • Reasons for risk

Entity detail panel:

PERSON Risk: HIGH Appears in: FIR_001, FIR_019, FIR_023 Connections: 12 Locations: 3 Vehicles: 2 Why flagged? ✓ Cross-FIR connections ✓ High graph centrality ✓ Shared identifier

Graph filters:

People • Locations • Vehicles • Phones • Organizations • FIRs

Performance target:

Do heavy preprocessing/indexing offline. During the live investigation, query the processed database/graph instead of OCR-processing the entire dataset after every click.

7. MEET — ARCHITECTURE + INTEGRATION + AI

Goal: make all six modules operate as one product and provide the AI explanation layer.

Integration flow:

Krishna UI → Anika API → Aayush Processing → Aayushman Intelligence → Database → Results → Krishna UI

AI role:

AI should explain and summarize evidence-backed findings rather than independently deciding who is criminal or generating an unsupported risk score.

Example:

System detection: Risk Score = 87/100 Risk Level = HIGH AI explanation: The entity was flagged because it has multiple cross-case connections, shares identifiers with other entities, and belongs to a high-density network cluster.

Meet should also maintain one fixed demo dataset and freeze the JSON/API contracts early so modules can be integrated without breaking changes.

8. SAFINA — CYBERSECURITY + QA + PERFORMANCE

Goal: ensure the system is secure, reliable and fast enough for the live demonstration.

Security tests:

Authentication • Authorization • Case-level access control • File-upload validation • Malicious filenames • Unsupported extensions • API validation • XSS • SQL injection • Broken access control • Secret exposure • CORS/configuration

Dataset tests:

Corrupted FIR • Empty FIR • Duplicate FIR • Duplicate OCR records • Missing text • Low-confidence OCR • Huge input • Malformed JSON • Unsupported file

Intelligence tests:

Verify that unrelated entities do not receive strong relationships; repeated identifiers create relationship candidates; cross-FIR appearances are captured; and every risk score has machine-readable reasons.

Performance measurements:

Dataset ingestion time • preprocessing time • entity extraction time • graph construction time • analysis time • API response time • dashboard load time

9. FINAL PERFORMANCE-OPTIMIZED ARCHITECTURE

544 FIR Images / 2,447 OCR Records ↓ OFFLINE DATA PROCESSING ↓ Clean + Normalize + Extract ↓ ENTITY RESOLUTION ↓ KNOWLEDGE GRAPH ↓ GRAPH + PATTERN ANALYSIS ↓ EXPLAINABLE RISK SCORE ↓ ANIKA BACKEND ↓ KRISHA INVESTIGATION UI ↓ MEET AI EXPLANATION ↓ INVESTIGATOR REPORT SAFINA: SECURITY + QA ACROSS EVERY STAGE

10. SIX-MEMBER RESPONSIBILITY TABLE

Member	Owns	Main Output
Aayush	Dataset + OCR + preprocessing + extraction	Structured FIR JSON
Aayushman	Graph + detection + scoring	Intelligence / analysis JSON
Anika	Backend + DB + APIs	Central data/API layer
Krishna	Dashboard + graph + investigation UI	Investigator interface
Meet	Architecture + integration + AI explanation + demo	Complete integrated system
Safina	Security + QA + performance	Validated and secure system

11. LIVE SIH DEMO FLOW

1. LOGIN 2. CREATE / OPEN CASE 3. LOAD FIR DATA 4. SYSTEM PROCESSES FIR EVIDENCE 5. ENTITIES ARE EXTRACTED 6. RELATIONSHIPS ARE DISCOVERED 7. NETWORK GRAPH IS BUILT 8. SUSPICIOUS PATTERN IS DETECTED 9. EXPLAINABLE RISK SCORE IS GENERATED 10. AI EXPLAINS WHY THE ENTITY/NETWORK WAS FLAGGED 11. ALERT IS GENERATED 12. INVESTIGATION REPORT IS DISPLAYED

12. KEY TEAM RULES

- Do not build six independent mini-projects.
- Freeze JSON/API interfaces early.
- Do not push directly to main; use feature branches and pull requests.
- Keep evidence traceability from every finding back to a FIR/evidence source.
- Do not rely on AI for the underlying criminal-risk decision.
- Prefer offline preprocessing and indexed graph/data queries for live-demo performance.
- Use the same processed demo dataset across all six modules.
- Every risk score should have explicit, inspectable reasons.

Final architecture principle: **Krishna UI** → **Anika API** → **Aayush Processing** → **Aayushman Intelligence** → **Meet AI/Integration** → **Safina Validation** → **back to the UI**.